Title: Creating DataFrames

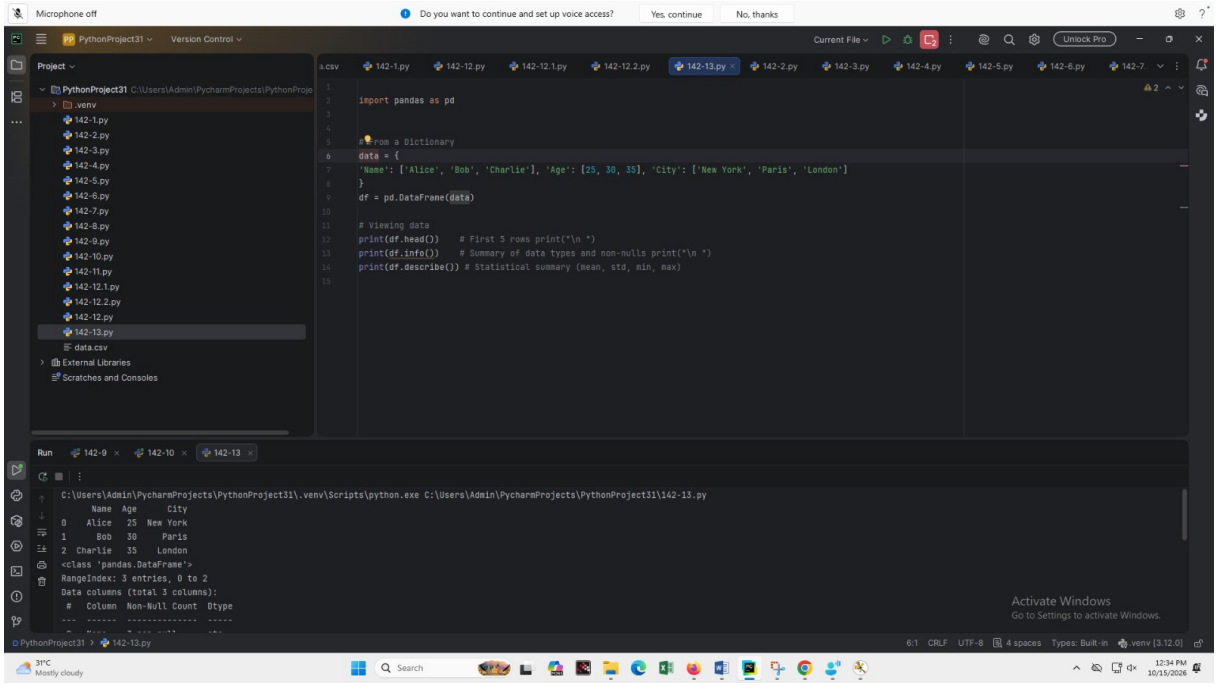
Aim:

To create a Pandas Series and DataFrame from lists, dictionaries and NumPy arrays.

Algorithm:

1. Start.
2. Import the Pandas library (`import pandas as pd`).
3. Create a Series using `pd.Series()`.
4. Create a DataFrame from a dictionary using `pd.DataFrame()`.
5. Display the Series and DataFrame.
6. Stop.

Program:



Title: Selection and Filtering

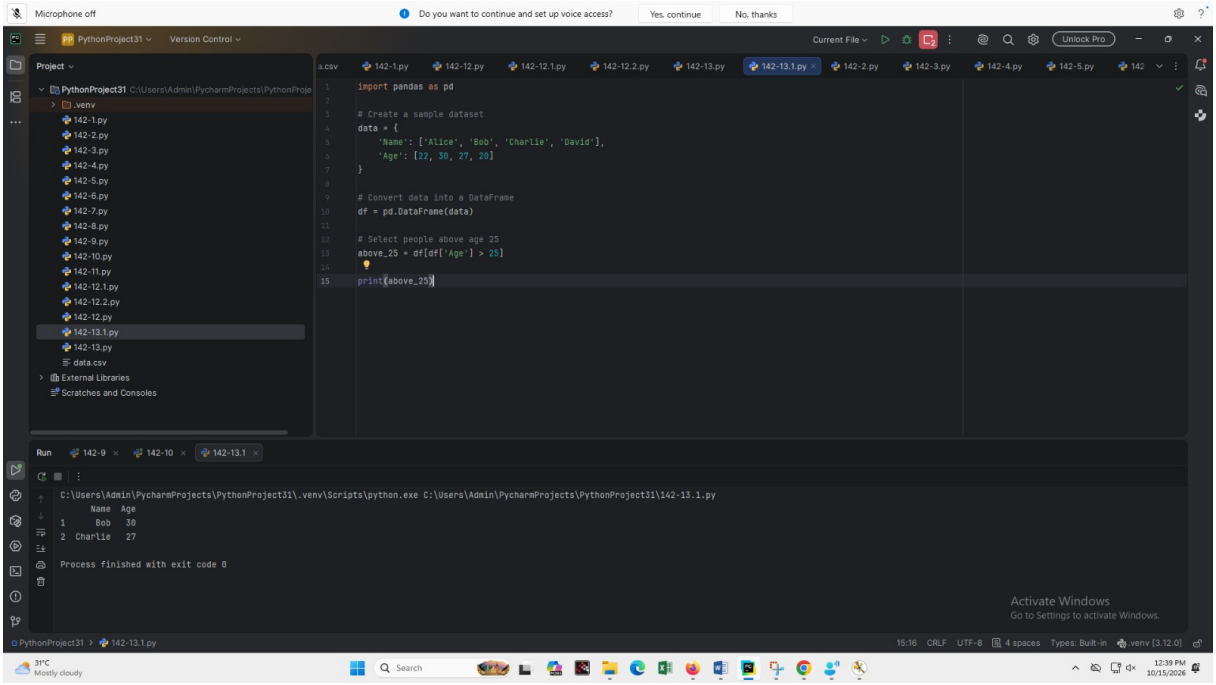
Aim:

To select specific rows and columns from a DataFrame using loc and iloc, and to filter rows based on one or more conditions.

Algorithm:

1. Start.
2. Import Pandas and create a DataFrame.
3. Select a column using `df["column"]` and multiple columns using `df[["col1", "col2"]]`.
4. Select rows by label using `df.loc[]`.
5. Select rows by position using `df.iloc[]`.
6. Define a condition (e.g., `df["Age"] > 25`).
7. Apply the condition to the DataFrame (`df[condition]`) and combine conditions using & (and) and | (or).
8. Display the selected and filtered results.
9. Stop.

Program:



Title: Handling Missing Values

Aim:

To detect, remove and fill missing values in a DataFrame.

Algorithm:

1. Start.
2. Import Pandas and NumPy, and create a DataFrame containing `np.nan` values.
3. Detect missing values using `isnull()` and `isnull().sum()`.
4. Remove rows with missing values using `dropna()`.
5. Fill missing values using `fillna()` (with a constant or the mean).
6. Display the results.
7. Stop.

Program:

